# 외계행성 대기
# Exoplanetary Atmospheres

## Lecture 7: Vertical Thermal Structure of a Planetary Atmosphere
## 2026 October 26 (Monday)

**Based on: Chapters 6 and 9 of Seager, *Exoplanet Atmospheres* (2010)**

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# Outline – Lecture 7

Approximate Solutions: Rosseland, Eddington, and the Gray Atmosphere

Why the Vertical Structure Matters

Hydrostatic Equilibrium and the Scale Height

Surface Temperature in Simple Greenhouse Models

Summary

# Approximate solutions

How to solve the radiative transfer equation:

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = -\alpha_\nu^{\mathrm{ext}}\,(I_\nu - S_\nu), \qquad S_\nu = (1-\varepsilon_\nu)J_\nu + \varepsilon_\nu B_\nu, \quad \text{with} \quad \varepsilon_\nu = \frac{\alpha_\nu^{\mathrm{abs}}}{\alpha_\nu^{\mathrm{ext}}}.$$

(Recall: $S_\nu \equiv \varepsilon_\nu^{\mathrm{tot}}/\alpha_\nu^{\mathrm{ext}}$, where the total emissivity is thermal emission, $\alpha_\nu^{\mathrm{abs}} B_\nu$ in LTE (Kirchhoff), plus scattering into the beam, $\alpha_\nu^{\mathrm{scat}} \oint \mathscr{P}(\Theta) I_\nu \,\mathrm{d}\Omega'$ with $\oint \mathscr{P}\,\mathrm{d}\Omega = 1$, which for isotropic ($\mathscr{P} = 1/4\pi$) and coherent scattering is $\alpha_\nu^{\mathrm{scat}} \oint I_\nu \,\mathrm{d}\Omega'/4\pi = \alpha_\nu^{\mathrm{scat}} J_\nu$ (Lecture 3); dividing by $\alpha_\nu^{\mathrm{ext}} = \alpha_\nu^{\mathrm{abs}} + \alpha_\nu^{\mathrm{scat}}$ gives the weights $\varepsilon_\nu$ and $1-\varepsilon_\nu$.)

(Here $\varepsilon_\nu = 1 - \omega_0$: the fraction of extinction that is true absorption, so $S = (1-\omega_0)B + \omega_0 J$ as before.)

We will learn two approximations to solve the equation in a **plane-parallel** medium, primarily developed for stellar atmospheres:

- the **Rosseland** approximation (radiative diffusion),
- the **Eddington** approximation.

Together they give the temperature structure of a *gray* atmosphere and the limb darkening of the Sun, and they lead to the Milne solution.

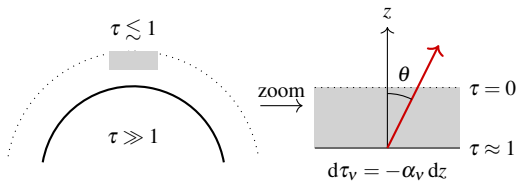
# Plane-parallel atmosphere

**Stellar (and planetary) atmosphere.**

- The atmosphere of a star (or planet) is its outer region, which determines the properties of the radiative flux that emanates from (are incident on) its surface.
- Computing its structure needs the detailed monochromatic opacity and powerful numerical codes; but with appropriate approximations interesting results follow analytically.

**Plane-parallel atmosphere.**

- We ignore the curvature of the underlying body (frequency independent opacity). Suitable when the atmosphere is a thin layer, "locally virtually flat".
- The optical depth is a function of the vertical distance $z$ rather than the distance $s$ along the ray; it increases downward, with $\tau = 0$ at the top: $\mathrm{d}\tau_\nu = -\alpha_\nu \, \mathrm{d}z$.



# (1) Radiative diffusion: the Rosseland approximation

Imagine a plane-parallel medium in which $\rho$ and $T$ depend only on the depth $z$. Along a ray, $\mathrm{d}s = \mathrm{d}z/\cos\theta = \mathrm{d}z/\mu$, so

$$\mu \frac{\partial I_\nu(z,\mu)}{\partial z} = -\alpha_\nu^{\mathrm{ext}}(I_\nu - S_\nu) \quad \Longrightarrow \quad I_\nu(z,\mu) = S_\nu - \frac{\mu}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial I_\nu}{\partial z}.$$

- **"Zeroth" approximation.** Deep in the material, *all quantities change slowly on the scale of a mean free path* $\ell = 1/\alpha^{\mathrm{ext}}$, so the derivative term is very small:

$$I_\nu^{(0)}(z,\mu) \approx S_\nu^{(0)}(T) \quad \therefore \quad J_\nu^{(0)} = S_\nu^{(0)} \text{ and } I_\nu^{(0)} = S_\nu^{(0)} = B_\nu \quad (\mathrm{LTE}),$$

  independent of the angle.

- **"First" approximation.** Insert $I^{(0)}$ in the derivative term:

$$I_\nu^{(1)}(z,\mu) \approx S_\nu^{(0)} - \frac{\mu}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial I_\nu^{(0)}}{\partial z} = B_\nu(T) - \frac{\mu}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu(T)}{\partial z} \quad \to \text{ linear in } \mu.$$

  With $\mathrm{d}\tau_\nu = -\alpha_\nu^{\mathrm{ext}}\mathrm{d}z$ this is $I_\nu \simeq B_\nu + \mu\,\mathrm{d}B_\nu/\mathrm{d}\tau_\nu$.

# What the first approximation means

$$\boxed{I_\nu(\tau_\nu,\mu) \simeq B_\nu(\tau_\nu) + \mu\,\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}} \qquad (\tau_\nu \text{ increasing inward})$$

- The leading term is **isotropic and Planckian**, $I_\nu \simeq S_\nu \simeq B_\nu$; as $\tau \to \infty$, $I = S = B$. A small anisotropy $\propto \mu$ carries the net flux.
- Its moments give $K = J/3$ *exactly* — the Eddington relation of the next part, and the **third Eddington coefficient** $\varepsilon_3 = K/J = 1/3$ (Heng's textbook). In the two-stream picture the up- and down-going streams merge here: net and total fluxes are no longer distinct, and scattering is effectively isotropic ($g_0 = 0$). (see next page)
- This is a **deep-interior** approximation, not a solution for the escaping radiation at $\tau_\nu \lesssim 1$.
- Large *scattering* depth alone does not guarantee thermalization: $S_\nu \to B_\nu$ requires enough true absorption ($\tau_* = \sqrt{3\varepsilon}\,\tau \gg 1$, below).

Seager's Eq. (6.49) prints $I \approx B - \mu\,\mathrm{d}B/\mathrm{d}\tau$; with $\tau$ increasing inward the sign must be $+$ to agree with the transfer equation and with her Eq. (6.51).

# Why the two streams merge deep inside

Deep inside, $I_\nu \simeq B_\nu + \mu\, \mathrm{d}B_\nu/\mathrm{d}\tau_\nu$ is almost isotropic. Two simplifications of the two-stream picture follow.

- **Net and total fluxes are no longer distinct.** The two hemispheric fluxes are

$$F_\pm = 2\pi \int_0^1 I_\nu(\pm\mu)\, \mu\, \mathrm{d}\mu \simeq \pi B_\nu \pm \frac{2\pi}{3} \frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}.$$

  Their sum, the total flux $\simeq 2\pi B_\nu$, is fixed by the local temperature and carries no new information. Only the small difference, the net flux $F_+ - F_-$ (from the $\mu$ term), matters. One flux and $J$ suffice; two separate streams are not needed.

- **Scattering is effectively isotropic.** If light arrives equally from all directions, it does not matter which way each scattering sends it: the scattered light is again the same in all directions, $\oint \mathscr{P} I\, \mathrm{d}\Omega' = J$ for any phase function. So the shape of the phase function (forward or backward scattering) hardly matters deep inside. Its only remaining effect, on the small net flux, is simply to change how effectively scattering blocks the flow (less for forward scattering), as if $\alpha^{\mathrm{scat}}$ were replaced by $(1 - g_0)\, \alpha^{\mathrm{scat}}$. We can therefore set $g_0 = 0$.

# The net flux in the first approximation

The angle-independent part of the intensity does not contribute to the net flux along $z$:

$$F_\nu(z) = \int I_\nu^{(1)}(z,\mu)\cos\theta\,\mathrm{d}\Omega = 2\pi\int_{-1}^{+1} I_\nu^{(1)}(z,\mu)\,\mu\,\mathrm{d}\mu$$
$$= -\frac{2\pi}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu}{\partial z}\int_{-1}^{+1}\mu^2\,\mathrm{d}\mu = -\frac{4\pi}{3\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu(T)}{\partial T}\frac{\partial T}{\partial z} = \frac{4\pi}{3}\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}.$$

Integrating over frequency, the **total flux** is

$$F(z) = \int_0^\infty F_\nu(z)\,\mathrm{d}\nu = -\frac{4\pi}{3}\frac{\partial T}{\partial z}\int_0^\infty \frac{1}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu.$$

Define the **Rosseland mean absorption coefficient**

$$\boxed{\frac{1}{\alpha_R} \equiv \frac{\displaystyle\int_0^\infty \frac{1}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu}{\displaystyle\int_0^\infty \frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu}}$$

# The equation of radiative diffusion

Use $\int_0^\infty \frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu = \frac{\partial}{\partial T}\int_0^\infty B_\nu\,\mathrm{d}\nu = \frac{\partial(\sigma T^4/\pi)}{\partial T} = \frac{4\sigma T^3}{\pi}$. Then the Rosseland approximation to the radiative flux is

$$\boxed{F(z) = -\frac{16\sigma T^3}{3\alpha_R}\frac{\partial T}{\partial z} \;\rightarrow\; -\chi\nabla T}$$

the **equation of radiative diffusion**, one of the fundamental equations for stellar interiors.

- It has the form of Fourier's law of heat conduction, with the **radiative conductivity** $\chi = 16\sigma T^3/3\alpha_R$: flux travels *down* the temperature gradient.
- **Per unit mass** (Heng): $\alpha_R = \rho\kappa_\mathrm{R}$. With hydrostatic equilibrium, $\mathrm{d}P = -\rho g\,\mathrm{d}z$,

$$F = \frac{16\,\sigma\,g\,T^3}{3\,\kappa_\mathrm{R}}\frac{\mathrm{d}T}{\mathrm{d}P}.$$

Notation: $\alpha$ is extinction per unit length — Seager's $\kappa$, Heng's $\alpha_e$; $\kappa_\mathrm{R} = \alpha_R/\rho$ is the Rosseland mean per unit mass.

# Why a harmonic mean

Three ways to average 1 and 100 (the Pythagorean means):

$$\text{arithmetic } \frac{1+100}{2} = 50.5, \qquad \text{geometric } (1\times 100)^{1/2} = 10, \qquad \text{harmonic } \frac{2}{1+1/100} \approx 2.$$

- The harmonic mean is dominated by the *smallest* value. The Rosseland mean averages $1/\alpha_\nu^{\mathrm{ext}}$, so the frequencies at which the extinction is **small** (transparent) dominate.
- In an optically thick medium radiation leaks out through these **windows**, just as current in parallel resistors flows through the smallest resistance.
- The weighting function $\partial B_\nu/\partial T$ resembles the Planck function but peaks at $h\nu_{\max} = 3.8\,k_{\mathrm{B}}T$ instead of $2.8\,k_{\mathrm{B}}T$.
- Hence a small amount of an opacity source that fills a window (a continuum absorber, a cloud) can change $\alpha_R$ and the deep thermal structure far more than a strong line in an already opaque band.

# (2) The Eddington approximation

In the Eddington approximation the intensities are assumed to approach *isotropy*, not necessarily their thermal values (in the Rosseland approximation they approach the Planck function at large depth). Near isotropy can be introduced by taking the intensity linear in $\mu$ (frequency suppressed):

$$I(\tau,\mu) = a(\tau) + b(\tau)\,\mu.$$

The first three moments are

$$\text{mean intensity:} \quad J \equiv \tfrac{1}{2}\int_{-1}^{+1} I\,\mathrm{d}\mu = a,$$

$$\text{flux:} \quad H \equiv \tfrac{1}{2}\int_{-1}^{+1} \mu I\,\mathrm{d}\mu = \frac{b}{3},$$

$$\text{radiation pressure:} \quad K \equiv \tfrac{1}{2}\int_{-1}^{+1} \mu^2 I\,\mathrm{d}\mu = \frac{a}{3},$$

so that

$$\boxed{K = \tfrac{1}{3}J} \qquad \text{the } \textbf{Eddington approximation}.$$

Compare the isotropic radiation field, $p = \frac{1}{3}u$, with $p \equiv \frac{1}{c}\int I\cos^2\theta\,\mathrm{d}\Omega$ and $u = \frac{4}{c}I$.

# The Eddington approximation: the radiative diffusion equation

The optical depth and the transfer equation are $\mathrm{d}\tau(z) \equiv -\alpha^{\mathrm{ext}}\mathrm{d}z$ and $\mu\,\partial I/\partial\tau = I - S$. The source function does not depend on $\mu$, because $S = (1-\varepsilon)J + \varepsilon B$. Integrating over $\mu$:

$$\tfrac{1}{2}\int_{-1}^{+1}\mathrm{d}\mu\left(\mu\frac{\partial I}{\partial\tau} = I - S\right) \;\rightarrow\; \frac{\partial H}{\partial\tau} = J - S, \qquad \tfrac{1}{2}\int_{-1}^{+1}\mathrm{d}\mu\,\mu\left(\mu\frac{\partial I}{\partial\tau} = I - S\right) \;\rightarrow\; \frac{\partial K}{\partial\tau} = H \;\rightarrow\; \frac{1}{3}\frac{\partial J}{\partial\tau} = H.$$

The two equations combine into

$$\boxed{\frac{1}{3}\frac{\partial^2 J}{\partial\tau^2} = J - S = \varepsilon\,(J - B)}$$

Define a new optical depth

$$\tau_* \equiv \sqrt{3\varepsilon}\,\tau = \sqrt{3\tau_{\mathrm{abs}}(\tau_{\mathrm{abs}} + \tau_{\mathrm{sca}})},$$

and the radiative equation becomes

$$\boxed{\frac{\partial^2 J}{\partial\tau_*^2} = J - B}$$

the **radiative diffusion equation**: given the temperature structure $B(\tau)$ of the medium, it can be solved for $J$.

# The gray atmosphere

If **the opacity is independent of frequency**, illustrative and insightful results can be found; a stellar atmosphere with such an opacity spectrum is a *gray atmosphere*.

- The radiation field still has a frequency distribution, since matter emits a spectrum determined by its temperature.
- We look for the temperature profile (versus depth) and the flux that emanates from a gray atmosphere.
- It is now more convenient to let $\tau$ increase *inward* (independent of frequency):

$$\boxed{\mu \frac{\mathrm{d}I_\nu}{\mathrm{d}\tau} = I_\nu - S_\nu}$$

- **(1)** Integrating over all frequencies,

$$\boxed{\mu \frac{\mathrm{d}I}{\mathrm{d}\tau} = I - S} \qquad I = \int_0^\infty I_\nu \,\mathrm{d}\nu, \quad S = \int_0^\infty S_\nu \,\mathrm{d}\nu.$$

# Radiative equilibrium

If there are no energy sources or sinks in the atmosphere, the energy absorbed by a layer equals the energy it emits. In the plane-parallel approximation this means the energy generated in the interior passes out as a *constant energy flux* through the outer layers.

$$\iint \alpha_\nu I_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega = \iint j_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega \qquad \text{(energy absorbed or scattered } = \text{ energy radiated)}$$

In the **gray atmosphere**, with the definition of the source function ($j_\nu = \alpha_\nu S_\nu$),

$$\iint \alpha_\nu I_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega = \iint \alpha_\nu S_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega \quad \rightarrow \quad \alpha \int J_\nu \,\mathrm{d}\nu = \alpha \int S_\nu \,\mathrm{d}\nu \quad \Longrightarrow \quad \boxed{J = S}$$

# Radiative equilibrium: the moment equations

**(2)** Average the first equation over the solid angle:

$$\tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu \left(\mu \frac{\mathrm{d}I_\nu}{\mathrm{d}\tau}\right) = \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, I_\nu - \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, S_\nu \;\Rightarrow\; \frac{\mathrm{d}H}{\mathrm{d}\tau} = J - S \xrightarrow{J=S} \boxed{\frac{\mathrm{d}H}{\mathrm{d}\tau} = 0, \quad H = \text{constant}}$$

**(3)** Multiply the first equation by $\mu$ and average over the solid angle:

$$\tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu \left(\mu^2 \frac{\mathrm{d}I}{\mathrm{d}\tau}\right) = \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, (\mu I) - \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, (\mu S) \;\Rightarrow\; \boxed{\frac{\mathrm{d}K}{\mathrm{d}\tau} = H} \quad \text{(isotropic source function).}$$

Integrating gives the radiation pressure as a function of optical depth,

$$\boxed{K(\tau) = H(\tau + c)} \qquad (Hc \text{ is the constant of integration}).$$

Using the **Eddington approximation**, $K = J/3$, the mean intensity is

$$\boxed{J(\tau) = 3H(\tau + c)} \qquad \text{with } c = \tfrac{2}{3}, \text{ shown on the next slide.}$$

# Derivation of the constant of integration

Solve $\mu\,\mathrm{d}I/\mathrm{d}\tau = I - S$ formally. For $\mu > 0$ (outgoing) multiply by $e^{-\tau/\mu}$ and integrate from $\tau' = \infty$ to $\tau$; for $\mu < 0$ (ingoing) integrate from $\tau' = 0$, where nothing comes in:

$$\mu > 0:\ I(\tau,\mu) = \int_{\tau}^{\infty} e^{-(\tau'-\tau)/\mu} S \frac{\mathrm{d}\tau'}{\mu}, \qquad \mu < 0:\ I(\tau,\mu) = \int_{0}^{\tau} e^{-(\tau-\tau')/\mu} S \frac{\mathrm{d}\tau'}{|\mu|}.$$

At the surface, for $\mu > 0$ and with $S = J = 3H(\tau + c)$,

$$I(0,\mu) = \int_0^{\infty} e^{-\tau'/\mu} S \frac{\mathrm{d}\tau'}{\mu} = 3H \int_0^{\infty} e^{-\tau'/\mu} (\tau' + c) \frac{\mathrm{d}\tau'}{\mu} \quad \Longrightarrow \quad \boxed{I(0,\mu) = 3H(\mu + c)}$$

and $I(0,\mu) = 0$ for $\mu < 0$. Now use the definition of the outward flux $H$ at $\tau = 0$:

$$H = \tfrac{1}{2}\int_{-1}^{1} \mu I(0,\mu)\,\mathrm{d}\mu = \tfrac{1}{2}\int_{0}^{1} \mu\, I(0,\mu)\,\mathrm{d}\mu = \frac{3H}{2}\left(\frac{1}{3} + \frac{c}{2}\right) \quad \Longrightarrow \quad \boxed{c = \frac{2}{3}}$$

# Limb darkening

- Observations show that the intensity at the center of the solar disk is larger than near its limb: **limb darkening**.
- The local temperature increases with depth, and the radiation leaving the star comes from $\tau \approx 1$ along the line of sight. **The layers responsible for the radiation are hotter at the center than near the limb.**
- The Eddington limb-darkening law can be estimated from the gray-atmosphere results (next slide).

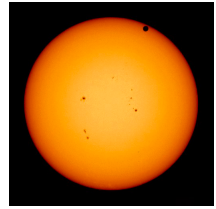
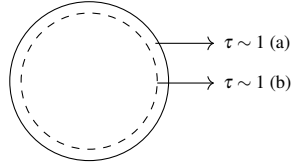


Image: Brocken Inaglory.




$T_a > T_b \Rightarrow S_a > S_b$

# Limb darkening: the Eddington law

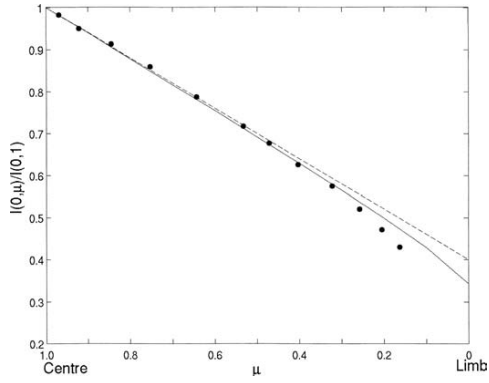
The intensity at the surface of the star is, with $S = J = 3H\,(\tau + \frac{2}{3})$,

$$I(\tau = 0, \mu) = \int_0^\infty S(\tau)\, e^{-\tau/\mu}\, \frac{\mathrm{d}\tau}{\mu}$$
$$= \int_0^\infty 3H\Big(\tau + \frac{2}{3}\Big) e^{-\tau/\mu}\, \mathrm{d}(\tau/\mu) = 3H\Big(\mu + \frac{2}{3}\Big).$$

The direction toward the center of the disk is $\theta = 0$, $\mu = 1$; the limb corresponds to $\mu < 1$. The ratio of the intensities is

$$\boxed{\frac{I(0,\mu)}{I(0,1)} = \frac{3}{5}\Big(\mu + \frac{2}{3}\Big)}$$

the **Eddington limb-darkening law**. At the limb ($\mu = 0$) the ratio is 0.4: the intensity of the gray-atmosphere approximation at the limb is 40% of its value at the center of the disk.




Dots: observation. Dashed: Eddington approximation. Solid: exact gray-atmosphere solution; $\lambda = 5485$ Å (Pierce et al. 1950).

# The temperature profile of a gray atmosphere

From the definition of the effective temperature,

$$H = \frac{1}{4\pi}\int \cos\theta\, I\, \mathrm{d}\Omega = \frac{1}{4\pi}F = \frac{1}{4\pi}\sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4.$$

The relation between the intensity and the outward flux, $F = \pi J$, gives

$$J(\tau) = \frac{1}{\pi}F(\tau) = \frac{1}{\pi}\sigma_{\mathrm{SB}} T^4(\tau),$$

where $T$ is the effective temperature at $\tau$. Combining with $J(\tau) = 3H\left(\tau + \frac{2}{3}\right)$,

$$T^4 = \frac{3}{4}T_{\mathrm{eff}}^4\left(\tau + \frac{2}{3}\right) \quad \Longrightarrow \quad \boxed{T = T_{\mathrm{eff}}\left[\frac{3}{4}\left(\tau + \frac{2}{3}\right)\right]^{1/4}}$$

This tells us how the temperature varies inside a gray atmosphere.

- At $\tau = 2/3$, $T = T_{\mathrm{eff}}$: the temperature at the photosphere is the effective temperature.
- The same form follows from radiative diffusion alone: **Milne's solution** (next slide).

# Milne's solution

Deep in the atmosphere (no external flux), the flux is carried outward by radiation and, in radiative equilibrium, is constant with depth; set it to the internal heat flux, $F = \sigma_{\mathrm{SB}} T_{\mathrm{int}}^4$. With the Rosseland optical depth $\mathrm{d}\tau_R = -\alpha_R \,\mathrm{d}z$, the diffusion equation reads

$$\sigma_{\mathrm{SB}} T_{\mathrm{int}}^4 = \frac{16\sigma_{\mathrm{SB}} T^3}{3} \frac{\mathrm{d}T}{\mathrm{d}\tau_R} = \frac{4\sigma_{\mathrm{SB}}}{3} \frac{\mathrm{d}T^4}{\mathrm{d}\tau_R},$$

and integrating over $\tau_R$,

$$\boxed{T^4 = \frac{3}{4} T_{\mathrm{int}}^4 \,(\tau_R + C)}$$

with $C \sim 1$ a constant of integration.

- This is **Milne's solution**, first derived for stars. Diffusion alone cannot determine $C$, because it does not hold near the surface. The Eddington approximation with the boundary condition of no incoming radiation gives $C = 2/3$: exactly the gray-atmosphere profile of the previous slide, with $T_{\mathrm{int}} = T_{\mathrm{eff}}$.
- At $\tau_R \approx 1$ the observer measures $T \approx T_{\mathrm{int}}$: for the Sun, $T_{\mathrm{int}} \approx 5800\,\mathrm{K}$ is the photospheric temperature. Temperature rises as $\tau_R^{1/4}$ with depth.
- In an irradiated planet this internal part is joined by the absorbed starlight; combining the two gives the temperature–pressure profiles of Lecture 8.

# Boundary conditions: the two-stream approximation

To solve the second-order equation for $J$ we need two boundary conditions. One way is the **two-stream approximation**: the whole radiation field is represented by radiation at just two angles, $\mu = \pm\mu_0$,

$$\boxed{I(\tau,\mu) = I^+(\tau)\,\delta(\mu-\mu_0) + I^-(\tau)\,\delta(\mu+\mu_0)}$$

The two terms are the outward and inward intensities. The three moments are

$$J = \tfrac{1}{2}\left(I^+ + I^-\right), \qquad H = \tfrac{1}{2}\mu_0\left(I^+ - I^-\right), \qquad K = \tfrac{1}{2}\mu_0^2\left(I^+ + I^-\right),$$

and $K = \frac{1}{3}J$ requires $\mu_0 = 1/\sqrt{3}$ ($\theta_0 = \cos^{-1}\mu_0 = 54.74°$).

1. Solve the first two for $I^\pm$: $I^+ + I^- = 2J$ and $I^+ - I^- = 2H/\mu_0$, so $I^\pm = J \pm H/\mu_0$.
2. The first moment of $\mu\,\partial I/\partial\tau = I - S$ (multiply by $\mu$ and average over angle) gives $\partial K/\partial\tau = H$; with $K = \frac{1}{3}J$, $H = \frac{1}{3}\,\partial J/\partial\tau$.
3. Hence $H/\mu_0 = \sqrt{3}\cdot\frac{1}{3}\,\partial J/\partial\tau = \frac{1}{\sqrt{3}}\,\partial J/\partial\tau$, and

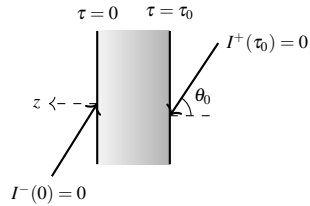
$$\boxed{I^\pm = J \pm \frac{1}{\sqrt{3}}\frac{\partial J}{\partial\tau}}$$

# Boundary conditions: a slab with no incident radiation

Let the medium extend from $\tau = 0$ to $\tau = \tau_0$ with no incident radiation, $I^-(0) = 0$ and $I^+(\tau_0) = 0$; with $I^{\pm} = J \pm \frac{1}{\sqrt{3}} \partial J / \partial \tau$ this gives

$$\boxed{\begin{aligned} \frac{1}{\sqrt{3}} \frac{\partial J}{\partial \tau} &= J \quad \text{at } \tau = 0, \\ \frac{1}{\sqrt{3}} \frac{\partial J}{\partial \tau} &= -J \quad \text{at } \tau = \tau_0. \end{aligned}}$$

Note the generalization to $N$ streams, in which the field is represented at $N$ angles.



# [Vertical Thermal Structure] The temperature–pressure (T–P) profile

The vertical temperature structure of an atmosphere is fundamental for several reasons.

- **Habitability.** The surface temperature and pressure decide whether liquid water is stable, and whether complex molecules can survive at all. It the surface is too hot for covalent bonds to form complex molecules, life cannot exist.
- **Chemistry.** The T–P structure is the starting point for equilibrium and non-equilibrium chemistry, i.e. for the composition.
- **Spectra.** As Lectures 5 and 6 showed, the emergent spectrum samples the temperature at the depth where each wavelength becomes optically thick; absorption or emission features depend on the *temperature gradient*.

Pressure is used as a proxy for altitude (the two are linked by hydrostatic equilibrium), hence the abbreviation **T–P profile**. There is no single equation for it: radiation, convection, gravity and the opacities all play a part.

This lecture treats the **one-dimensional** vertical structure. Hot, tidally locked exoplanets also have strong *horizontal* contrasts (permanent day and night sides); horizontal heat transport is the subject of atmospheric circulation.

# Solar System profiles look alike

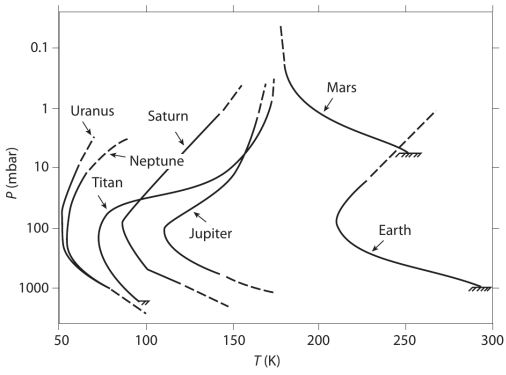



Figure 1: Vertical temperature structure of Solar System planet atmospheres. Seager (2010), Fig. 9.1, after Goody & Yung.

- Every profile has a **troposphere** where temperature falls with height, a **tropopause** minimum near 0.1 bar, and in most cases a **stratosphere** where it rises again.
- The thermal inversion comes from absorption of stellar ultraviolet: by **$CH_4$-induced hazes** in the giant planets and Titan, by **$O_3$** on Earth. Mars, with neither, has no stratospheric inversion.
- The similarity is not a coincidence: the same physics — hydrostatic equilibrium, radiative transfer, convection — shapes all of them (Seager's exercise 8). The aim of this lecture is to understand that physics.

# Earth's vertical structure

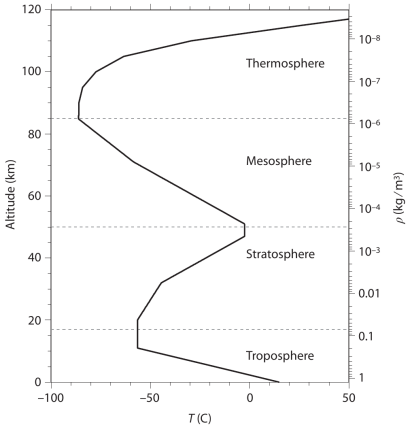



Figure 2: Earth's atmosphere at mid-latitudes (1976 U.S. Standard Atmosphere). Seager (2010), Fig. 9.2.

The layers are named by where the temperature gradient reverses; the reversals come from *where* solar radiation is absorbed, which depends on wavelength and composition.

- **Troposphere** (0–∼12 km): where we live and weather happens; holds ∼ 85% of the mass, because density falls exponentially with height. Most strong visible and infrared spectral features form here. Mt. Everest (8.8 km) does not reach the top.
- Temperature is highest at the ground: visible sunlight reaches the surface, which re-radiates in the infrared and heats the air from below (plus the greenhouse effect).
- **Stratosphere**: temperature *increases* with height, because $O_3$ absorbs solar UV. This also shields life from UV.
- **Mesosphere** and **thermosphere** matter mainly through photochemistry and atmospheric escape.

# Where sunlight is absorbed

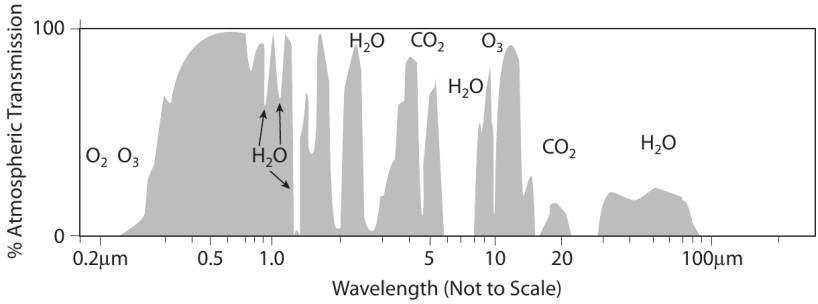



Figure 3: Transmission of solar radiation through Earth's atmosphere, with the main absorbers. Seager (2010), Fig. 9.3.

- The visible window ($\sim 0.3$–$1\,\mu$m) is nearly transparent: most sunlight reaches and heats the **ground**.
- $O_2$ and $O_3$ absorb the ultraviolet **high up**, heating the stratosphere and thermosphere.
- In the infrared, $H_2O$ and $CO_2$ make the atmosphere largely opaque, except in a few windows (e.g. 8–12 $\mu$m): the surface emission is absorbed and re-emitted by the air.

# Hydrostatic equilibrium

Gravity compresses the atmosphere; the gas pressure resists. Without the pressure force the atmosphere would collapse into a thin shell, without gravity it would diffuse into space. **"Static" means no vertical acceleration.**

For a slab of area $\delta A$ and thickness $\delta z$ (height $z$ upward), the pressure difference between its faces, $\mathrm{d}P = (\mathrm{d}P/\mathrm{d}z)\,\delta z$, must hold up its weight $\rho g\,\delta z\,\delta A$:

$$\frac{\mathrm{d}P}{\mathrm{d}z}\,\delta z\,\delta A = -g\rho\,\delta z\,\delta A \qquad \Longrightarrow \qquad \boxed{\frac{\mathrm{d}P}{\mathrm{d}z} = -g\rho}$$

In terms of the optical depth, $\mathrm{d}\tau = -\kappa\,\mathrm{d}z$ with $\kappa$ the (mean) extinction per unit length, or per unit mass $\kappa_m = \kappa/\rho$:

$$\frac{\mathrm{d}P}{\mathrm{d}\tau} = \frac{g\rho}{\kappa} = \frac{g}{\kappa_m} \qquad \Longrightarrow \qquad \tau = \frac{\kappa_m P}{g} \text{ for constant } \kappa_m.$$

- Valid almost everywhere in a planetary atmosphere. Exceptions have real vertical accelerations: hydrodynamic escape in the upper atmosphere, and small-scale vigorous systems (convective plumes, thunderstorms, tornadoes).

# Equation of state and the pressure scale height

For planetary atmospheres the ideal gas law is adequate,

$$P = nkT = \frac{\rho kT}{\mu_m m_{\rm H}}, \qquad \mu_m \equiv \frac{\bar{m}}{m_{\rm H}} \quad \text{(mean molecular weight)}.$$

Combining with hydrostatic equilibrium,

$$\frac{{\rm d}P}{P} = -\frac{\mu_m m_{\rm H} g}{kT}\,{\rm d}z \quad \Longrightarrow \quad \boxed{P = P_0\,{\rm e}^{-z/H}, \qquad H \equiv \frac{kT}{\mu_m m_{\rm H} g}}$$

for constant $T$, $\mu_m$ and $g$. $H$ is the height over which pressure (and density) fall by a factor e.

- Temperature varies by only a factor of a few through an atmosphere, so $P$ and $\rho$ still fall roughly exponentially.
- The observable spectrum comes from a region a few scale heights thick (roughly $5H$).

# How large is $H$? (Seager, exercise 2)

| planet | $T$ (K) | $g$ (m s$^{-2}$) | $\mu_m$ | $H$ (km) |
|---|---|---|---|---|
| Earth ($N_2$/$O_2$) | 288 | 9.8 | 29 | 8.4 |
| hot Jupiter ($H_2$/He) | 1500 | 25 | 2.3 | 220 |
| hot super-Earth, $H_2$-rich | 1000 | 20 | 2.3 | 180 |
| hot super-Earth, $CO_2$-rich | 1000 | 20 | 44 | 9.4 |

- $H \propto T/(\mu_m g)$: hot, low-gravity planets with light (hydrogen) atmospheres are the puffiest, i.e. the most vertically extended (large $H$ relative to the planet's radius).
- The **mean molecular weight** matters as much as the temperature: the same hot super-Earth has a 20 times larger $H$ if its atmosphere is $H_2$-rich. Since transmission features scale as $H$, this is how transit spectra distinguish hydrogen-rich from heavy atmospheres.
- Solar System values are 5–20 km; hot Jupiters reach hundreds of kilometers.

# Surface Temperature and Emission Temperature

Recall the energy balance (Lecture 4): with uniform redistribution of the absorbed starlight (the absorbed stellar radiation is circulated evenly around the planet), the energy emitted by the planet equals the absorbed stellar energy:

$$4\pi R_p^2 F_{S,p} = (1-A_\mathrm{B})\, F_{S,*} \left(\frac{R_*}{a}\right)^2 \pi R_p^2 \quad \Longrightarrow \quad \sigma T_\mathrm{eq}^4 = \frac{1}{4}\,\sigma T_*^4 \left(\frac{R_*}{a}\right)^2 (1-A_\mathrm{B}) \equiv \mathscr{F}_* \downarrow$$

$$\Longrightarrow \quad T_\mathrm{eq} = T_* \left(\frac{R_*}{2a}\right)^{1/2} (1-A_\mathrm{B})^{1/4}.$$

- $T_\mathrm{eq}$ is the temperature of an isothermal planet in complete equilibrium with its star. In an idealized atmosphere it is the temperature of the layer from which most of the radiation escapes to space: the **emission temperature** or **equilibrium temperature** $T_e \equiv T_\mathrm{eq}$.
- Question: how much warmer than $T_e$ is the *surface*? We answer it with one-layer models (after Marshall & Plumb), assuming
  - the star emits mostly in the visible, and all unscattered starlight reaches and heats the ground;
  - the atmosphere absorbs and emits only in the infrared.

# A simple greenhouse atmosphere: the setup

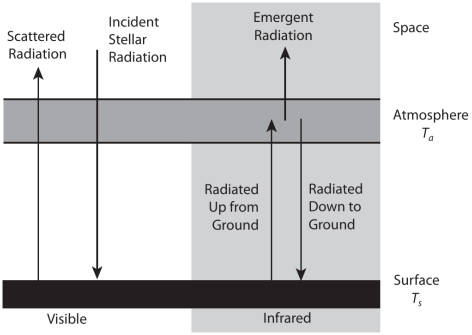



Figure 4: The simple greenhouse: one infrared-opaque layer at $T_a$ above a surface at $T_s$. Seager (2010), Fig. 9.4.

Two layers, still with uniform redistribution of the absorbed starlight (Seager §9.4, after Marshall & Plumb):

- a **surface** at temperature $T_s$, and above it
- an **atmosphere layer** at temperature $T_a$.

Starlight comes in at visible wavelengths and is absorbed by the ground; the ground and the layer exchange energy in the infrared.

Goal: $T_s$ and $T_a$ in terms of the emission temperature $T_e$.

# A simple greenhouse atmosphere: assumptions

- The star emits mostly in the **visible**. Apart from the part reflected to space (the Bond albedo $A_{\mathrm{B}}$), all starlight passes the layer and is **absorbed by the ground**: no scattering, no visible absorption in the layer.
- The heated ground re-emits in the **infrared**, at wavelengths set by its temperature.
- The layer is made of molecules that absorb and emit only in the infrared, and it is **opaque** there: it absorbs all the upwelling surface emission.
- The layer radiates as a black body at $T_a$, *both* upward to space and downward to the ground.

The visible/infrared separation is reasonable for Sun-like stars, whose output peaks in the visible, and for atmospheres of molecules ($H_2O$, $CO_2$, ...) that absorb and emit mainly in the infrared.

# A simple greenhouse atmosphere: energy balance

**Top of the atmosphere.** Only the layer is seen from space. From energy balance, its upward emission must carry away the absorbed starlight (Seager Eq. 9.16):

$$\sigma T_a^4 = \sigma T_*^4 \left(\frac{R_*}{2a}\right)^2 (1 - A_\mathrm{B}) = \sigma T_e^4 \quad \Longrightarrow \quad T_a = T_e.$$

The layer emitting to space is at the emission temperature, as it should be.

**Surface.** The ground absorbs the starlight *and* the downward emission from the atmosphere layer, and re-radiates both (Seager Eq. 9.18):

$$\sigma T_s^4 = \underbrace{\sigma T_*^4 \left(\frac{R_*}{2a}\right)^2 (1 - A_\mathrm{B})}_{\sigma T_e^4} + \underbrace{\sigma T_a^4}_{\sigma T_e^4} = 2\sigma T_e^4 \quad \Longrightarrow \quad \boxed{T_s = 2^{1/4} T_e \approx 1.19\, T_e}$$

**Check, the atmosphere layer.** It absorbs $\sigma T_s^4$ from below and emits $\sigma T_a^4$ upward plus $\sigma T_a^4$ downward: $\sigma T_s^4 = 2\sigma T_a^4$, the same result.

# A simple greenhouse atmosphere: Earth and Venus

$$T_s = 2^{1/4}\, T_e \approx 1.19\, T_e, \qquad T_a = T_e.$$

- The ground receives twice the stellar heating: the atmosphere layer contributes to heating the planet surface and "returns" half of the surface emission. This back-radiation is the greenhouse effect.
- Earth: $T_e \simeq 255\,\mathrm{K}$ gives $T_s \simeq 303\,\mathrm{K}$, but the observed mean is $\simeq 288\,\mathrm{K}$: **too warm** (the real atmosphere is not fully opaque in the infrared).
- Venus: $T_e \simeq 230\,\mathrm{K}$ gives $T_s \simeq 274\,\mathrm{K}$, but the observed $T_s \simeq 730\,\mathrm{K}$: **far too cold** ($(730/230)^4 \approx 100$; see "Many layers").
- The model overestimates Earth and badly underestimates Venus: a single, fully opaque layer is both too strong and too weak. Next we relax the opacity (leaky layer), then add layers.
- In the simple greenhouse model, we have assumed that the absorbed stellar radiation is trapped by the atmospheric layer and does not escape to space. In reality, some of the absorbed stellar radiation is re-emitted to space, which reduces the surface temperature.

# Take-home messages

**Approximate solutions.** Rosseland (deep layers): $F = -(16\sigma T^3/3\alpha_R)\,\mathrm{d}T/\mathrm{d}z$ with the harmonic mean $\alpha_R$, which with constant flux gives Milne's $T^4 = \frac{3}{4}T_{\mathrm{int}}^4(\tau_R + C)$. Eddington: $K = J/3$, which in a gray atmosphere gives $J = 3H(\tau + \frac{2}{3})$, $T^4 = \frac{3}{4}T_{\mathrm{eff}}^4(\tau + \frac{2}{3})$ and the limb-darkening law $I(\mu)/I(1) = \frac{3}{5}(\mu + \frac{2}{3})$.

**Structure.** Solar System atmospheres share a troposphere, tropopause near 0.1 bar and (UV-heated) stratosphere. Hydrostatic equilibrium, $\mathrm{d}P/\mathrm{d}z = -\rho g$, with the ideal gas law gives $P = P_0\mathrm{e}^{-z/H}$, $H = kT/\mu_m m_{\mathrm{H}} g$: 8 km for Earth, hundreds of km for hot Jupiters; $\mu_m$ matters as much as $T$.

**Simple greenhouse model.** One fully opaque infrared layer: $T_a = T_e$ and $T_s = 2^{1/4}T_e$. Atmospheric back-radiation warms the surface above the emission temperature. This model overestimates Earth's surface temperature and underestimates Venus's.

**Next (Lecture 8).** Leaky and multilayer greenhouse models, convection versus radiation, irradiated T–P profiles, the adiabat, and temperature retrieval.

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